Titer Determination of HCl 0.1 mol/L

Method for the standardization of 0.1 mol/L hydrochloric acid using Tris(hydroxymethyl)aminomethane (TRIS, or also THAM) as the primary standard. The titration is monitored with a combined pH glass electrode.

Sample	Primary standard Tris(hydroxymethyl) aminomethane (TRIS/THAM) 50 – 100 mg
Compound	Tris(hydroxymethyl)aminomethane, TRIS/THAM, $(HOCH_2)_3CNH_2$ M = 121.14 g/moL, z = 1
Chemicals	38% w/w Hydrochloric acid 50 mL Deionized water
Titrant	Hydrochloric acid, HCl c(HCl) = 0.1 mol/L
Standard	--
Indication	DGi111-SC combined pH glass electrode
Chemistry	$HCl + (HOCH_2)_3CNH_2 \rightarrow (OHCH_2)_3CNH_3^+ + Cl^-$
Calculation	R = m/(VEQ*c*C) C=M/(10*p*z) VEQ=Titrant consumption in mL c = Concentration of the titrant. m = mass of the sample in g. M = Molar mass of the sample p = Purity of the standard. z = Equivalent number
Waste disposal	Neutralization before final disposal as aqueous solution
Author, Version	Vineesh Pallath, IMSG AnaChem, Version 1.0 Revised: C. De Caro, MSG AnaChem

Preparation and Procedures

CAUTION

- Use safety goggles, a lab coat and wear gloves.
- Ensure accurate cleaning of sensor is sufficient after each titration.

<u>Preparation of 0.1 mol/L HCl:</u>

- Carefully add 8.5 mL of 38% w/w HCl to 1 L volumetric flask containing 500 mL of deionized water.
- Heat will generate; cool to room temperature and fill up to 1000mL with deionized water.

<u>Preparation of sample solution:</u>

- Weigh 0.05 – 0.10 g TRIS in to a glass titration beaker on the sample changer rack.
- Start the method.
- 50 mL of deionized water will be added by membrane pump of sample changer.
- The solution is stirred for 120 s and titration will proceed.
- After completion of titration, the electrode, stirrer and titration tubes will be rinsed by deionized water using the membrane pump.
- Subsequently, the electrode is conditioned for 30 s with deionized water in condition beaker.

Remarks

- The method parameters have been optimized for the sample of this application. It may be necessary to adapt the method to your specific sample.
- Glass beakers are recommended in order to avoid any interference during weighing due to electrostatic force.
- This method allows a fully automated analysis procedure. The method can be easily modified for manual operation. Select “Manual stand” in the method function “Titration stand”.

Instruments	
	- Titration Excellence T50/T70/T90 (Other Titrators: With modifications or manual operations are necessary) - Rondo 20 sample Changer with PowerShower™ - (ME-51108003) - XP205 Balance (ME-11106024)
Accessories	- 1 x 10mL DV1010 glass burette (ME-51107501) - Glass titration beaker (ME-00101446) - LabX® pro titration software

Results

All results

	Method-ID	TiterHCl
	Sample	0.06463 g (1/6)
	R1 (Titer)	1.00306
	Sample	0.06493 g (2/6)
	R1 (Titer)	1.00556
	Sample	0.06604 g (3/6)
	R1 (Titer)	1.00684
	Sample	0.06581 g (4/6)
	R1 (Titer)	1.00426
	Sample	0.06725 g (5/6)
	R1 (Titer)	1.00666
	Sample	0.06784 g (6/6)
	R1 (Titer)	1.00540

Statistics

	Method-ID	VPTiterHCl
	R1	Titer
	Samples	6
	Mean	1.00536
	s	0.00146
	srel	0.145 %

Titration curve

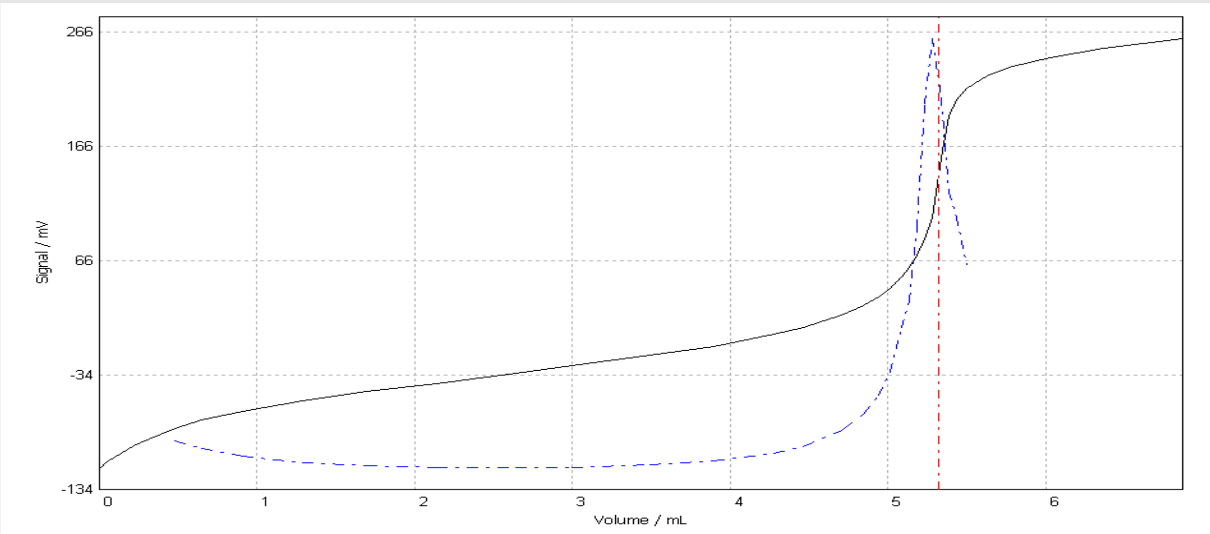

Table of measured values

	Volume mL	Increment mL	Signal mV	Change mV	1st deriv. mV/mL	Time s	Temperature °C
	0.0000	NaN	-115.3	NaN	NaN	0	25.0
	0.0500	0.0500	-109.9	5.4	NaN	3	25.0
	0.1000	0.0500	-105.7	4.2	NaN	6	25.0
	0.2175	0.1175	-95.8	9.9	NaN	9	25.0
	0.3140	0.0965	-89.5	6.2	NaN	12	25.0
	0.4775	0.1635	-80.6	9.0	51.00	16	25.0
	0.6490	0.1715	-73.6	7.0	41.35	19	25.0
	0.9110	0.2620	-65.9	7.7	31.28	22	25.0
	1.2910	0.3800	-56.7	9.2	23.34	25	25.0
	1.6900	0.3990	-48.4	8.3	19.11	28	25.0
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	3.4935	0.0500	-16.8	9.2	20.41	40	25.0
	3.8965	0.4030	-8.8	8.0	25.85	43	25.0
	4.2740	0.3775	2.1	10.9	35.17	46	25.0
	4.4625	0.1885	7.9	5.8	43.67	49	25.0
	4.7100	0.2475	18.3	10.4	63.88	53	25.0
	4.8480	0.1380	26.8	8.5	85.43	56	25.0
	4.9375	0.0895	33.8	7.0	109.33	59	25.0
	5.0180	0.0805	41.6	7.8	138.95	62	25.0
	5.0850	0.0670	51.6	10.0	195.24	65	25.0
	5.1350	0.0500	60.1	8.5	229.30	68	25.0
	5.1850	0.0500	71.2	11.1	322.38	73	25.0
	5.2350	0.0500	85.4	14.2	479.60	77	25.0
	5.2850	0.0500	103.5	18.1	562.76	80	25.0
EQP1	5.318853	NaN	138.7	NaN	571.15	NaN	NaN
	5.3350	0.0500	155.5	52.0	487.13	88	25.0
	5.3850	0.0500	192.2	36.7	367.62	92	25.0
	5.4350	0.0500	206.8	14.6	334.85	95	25.0
	5.5040	0.0690	216.2	9.4	273.37	98	25.0
	5.6290	0.1250	227.3	11.1	NaN	101	25.0
	5.7690	0.1400	234.6	7.3	NaN	104	25.0
	6.0260	0.2570	243.6	9.0	NaN	107	25.0
	6.3705	0.3445	251.7	8.1	NaN	110	25.0
	6.8695	0.4990	259.8	8.1	NaN	113	25.0

Comments

- Rondo sample changer with diaphragm pump was used. The conditioning time was set to 30 s along with rinsing was defined to 10mL. In this way, the electrode is cleaned with deionized water before starting the subsequent sample.
- The mean value of the titer is automatically stored as part of the setup by the function TITER.
- The purity of primary standard, TRIS, used is p = 100.00%

Literature:

Mettler-Toledo Application M003 (V1.0)

Mettler-Toledo Application M522 (V1.0)

Method

001 Title

Type	Titer determination
Compatible with	T50 / T70 / T90
ID	Titer HCl
Title	Titer 0.1 mol/L HCl
Author	admin
...	...

002 Sample (Titer)

Titrant	HCl
Concentration	0.1 mol/L
Standard	TRIS
Type of standard	Solid
Entry type	Weight
Lower limit	0.05 g
Upper limit	0.10 g
Density	1.0 g/mL
Correction factor	1.0
Temperature	25.0°C
Entry	Arbitrary

003 Titration stand (Rondo/Tower A)

Type	Rondo/Tower A
Titration stand	Rondo60/1A
Lid Handling	No

004 Pump

Auxiliary reagent	Water
Volume	50 mL
Condition	No

005 Stir

Speed	30 %
Duration	120 s
Condition	No

006 Titration (EQP) [1]

Titrant	
Titrant	HCl
Concentration	0.1 mol/L
Sensor	
Type	pH
Sensor	DG111-SC
Unit	mV
Temperature acquisition	
Temperature measurement	No
Stir	
Speed	30%
Predispense	
Mode	None
Wait time	0 s
Control	
Control	User
Titrant addition	Dynamic
dE (set value)	8.0 mV
dV (min)	0.05 mL
dV (max)	0.5 mL
Mode	Equilibrium controlled
dE	1.0 mV
dt	1 s
t (min)	3 s
t (max)	30 s
Evaluation and recognition	
Procedure	Standard
Threshold	200.0 mV/mL
Tendency	Positive
Ranges	0
Add. EQP criteria	No
Termination	
At Vmax	10.0 mL
At potential	No
At slope	No
After number of recognized EQPs	Yes
Number of EQPs	1
Combined termination criteria	No
Accompanying stating	
Accompanying stating	No
Condition	
Condition	No

007 Calculation R1

Result	Titer
Result unit	--
Formula	R1= m/(VEQ*c*C)
Constant	C= M/(10*p*z)
M	M[TRIS]
z	z[TRIS]
Decimal places	5
Result limits	No
Record statistics	Yes
Extra statistical func.	No
Send to buffer	No
Condition	No

008 Rinse

Auxiliary reagent	Water
Rinse cycles	1
Vol.per cycle	10 mL
Position	Current Position
Drain	No
Condition	No

009 Conditioning

Type	Fix
Interval	1
Position	Conditioning beaker
Time	30 s
Speed	60%
Condition	No

010 End of sample

011 Titer

Titrant	HCl
Concentration	0.1 mol/L
TITER	Mean[R1]
Limits	No
Condition	No

012 Calculation R2

Result	Mean Titer
Result unit	--
Formula	R2= Mean[R1]
Constant	C=1
M	M[TRIS]
z	z[TRIS]
Decimal places	5
Result limits	No
Record statistics	Yes
Send to buffer	No
Condition	No